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Advanced AI Training

Retail and eCommerce – Recommendation Engine for Personalized Shopping

Course Objectives

- Gain a comprehensive understanding of the current AI landscape in retail and eCommerce, including key recommendation technologies and trends.
- Analyze the current adoption of recommendation engines in global and Vietnamese retail, identifying both opportunities and challenges.
- Explore the provided workshop dataset, understanding its structure, relevance, and use in building personalized shopping experiences.
- Clarify the requirements and expectations for the group project, ensuring alignment with personalized recommendation goals.
- Develop skills to design and implement AI-driven recommendation systems tailored to customer preferences and behaviors.
- Evaluate ethical considerations and responsible AI practices in personalization, such as data privacy and fairness.

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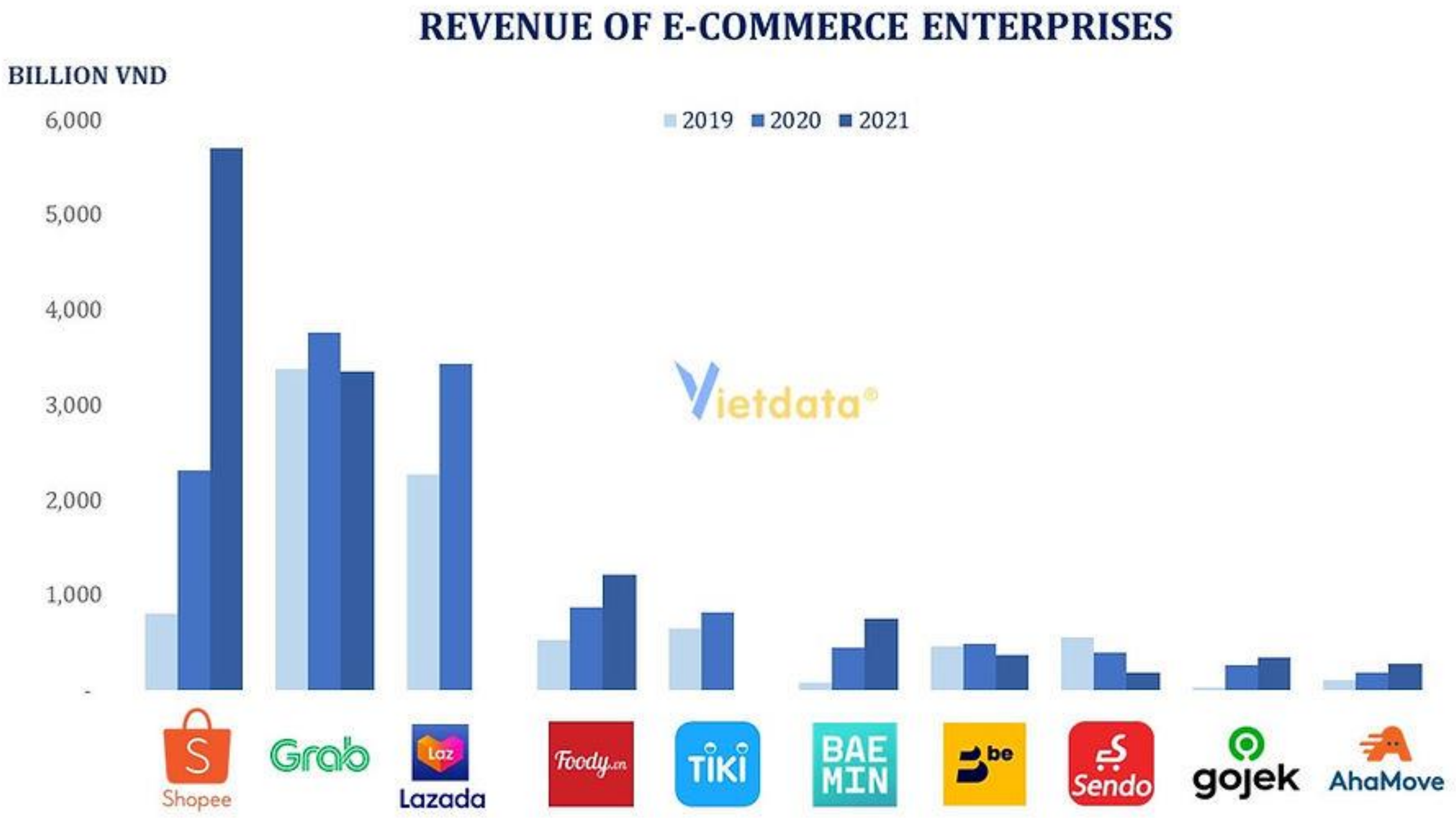
Overview of AI in Retail and eCommerce

ECommerce in Vietnam



AI Applications in Retail and eCommerce

AI in Retail and Ecommerce Use Cases

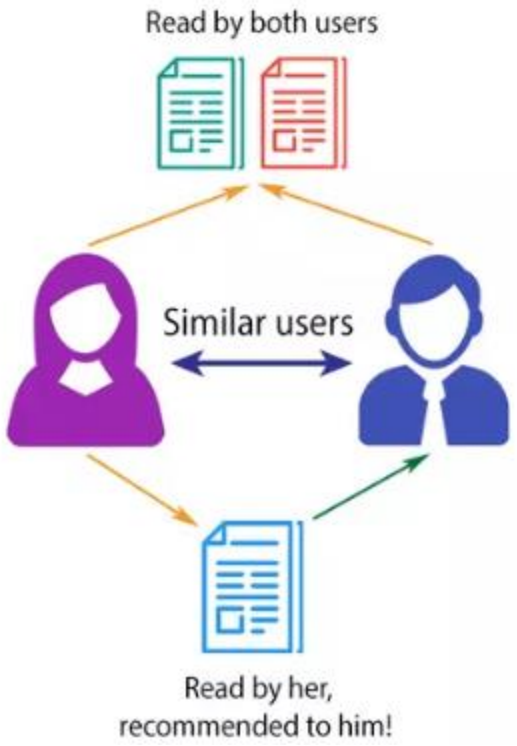
1. Personalized Discounts & Offers
2. Social & Influencer-Driven Recommendations
3. Personalized Product Recommendations
4. Dynamic Homepage & Category Personalization

Case Study: Personalized Product Recommendations

Tracks user actions: read history, searches, likes/dislikes, skips, and completion rate.

Tags book with metadata (genre, duration, topic, tone) to recommend similar content.

Uses collaborative filtering to suggest books liked by users with similar behavior.



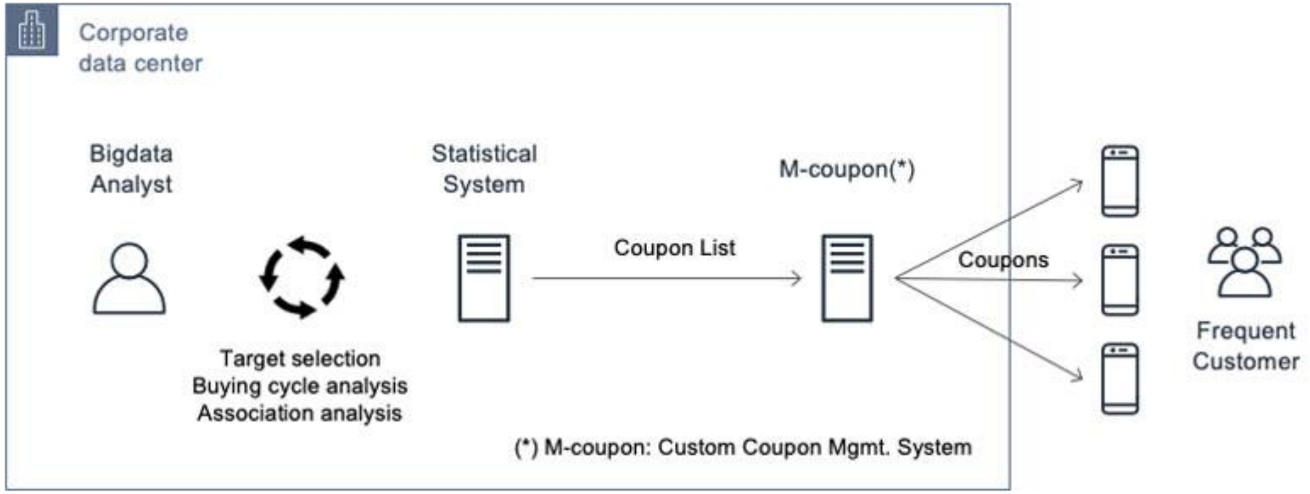
Case Study: Personalized Discounts & Offers

The system segments users by analyzing their browsing history, purchase frequency, product categories, and cart abandonment.

Custom discounts are generated in real time based on user's likelihood to convert.

Uses content-based filtering to offers for products similar to those a user viewed, even if not purchased.

Lotte Mart increased new product purchases by 1.7x. [Link here]

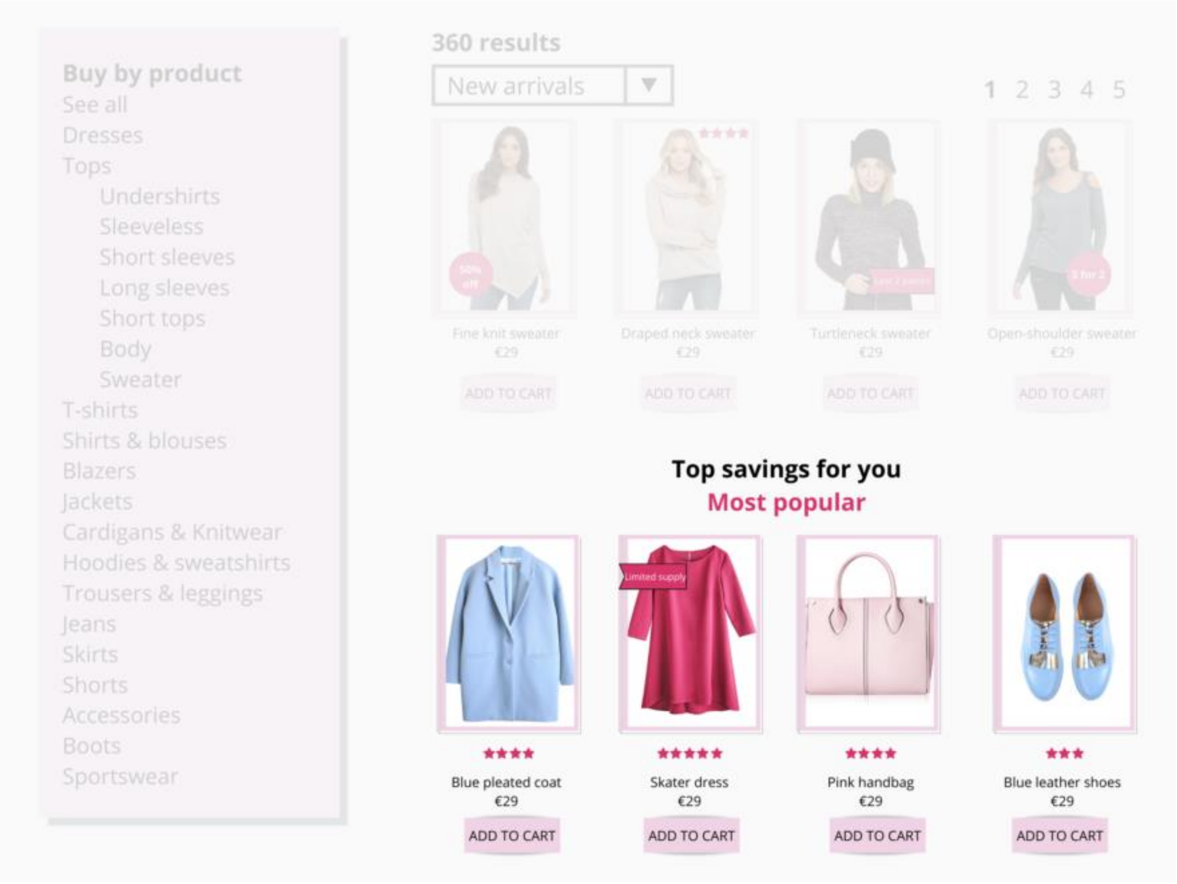


Case Study: Dynamic Homepage & Category Personalization

The system segments users based on their browsing history, purchase behavior, and category preferences.

Homepage layout and category sections are dynamically personalized in real time to match individual user interests.

Uses content-based filtering to offers for products similar to those a user viewed, even if not purchased.



Group Project Guideline

Overview

- This project aims to provide learners with hands-on experience in developing an AI-powered recommendation system for retail and eCommerce.

- By utilizing publicly available datasets, learners will explore the full AI development pipeline—from data collection, preprocessing, and feature engineering, to designing, training, and evaluating recommendation algorithms such as collaborative filtering or neural network-based models.

- The project will include performance evaluation using metrics like precision, recall, F1-score, and hit rate, and culminate in deploying the recommendation engine through a user-friendly interface or eCommerce-ready application.

- The ultimate goal is to build a practical system that delivers personalized product suggestions, improves customer satisfaction, and drives sales through data-driven personalization.

Datasets

Learners may either **use a dataset of their choice** or select from one of the recommended datasets listed below:

Retailrocket recommender system	Amazon Review	eCommerce behavior data from multi category store	User behavior data on Taobao App	MovieLens dataset
Includes user behavior data (clicks, add to cart, transactions) along with product and category information.	This is a large-scale Amazon Reviews dataset	This dataset contains 285 mil users' events from eCommerce website	This dataset is provided by Alibaba Group	MovieLens with 20 mil movie ratings

Dataset

Retailrocket recommender system **dataset**

https://www.kaggle.com/datasets/retailrocket/ecommerce-dataset/

Retailrocket provides structured data for building product recommendation systems based on user interactions like views and purchases

Table data (3 types of events, namely “view”, “addtocart” and “transaction”)

Descriptive | **Diagnostic** | **Predictive** | **Prescriptive**

Suggest next products based on customer behavior

Dataset

Amazon Review **dataset**

https://www.kaggle.com/datasets/bittlingmayer/amazonreviews/data/

A large-scale dataset of customer reviews collected from Amazon, covering a wide range of product categories. It includes star ratings, review text, product metadata, and user IDs

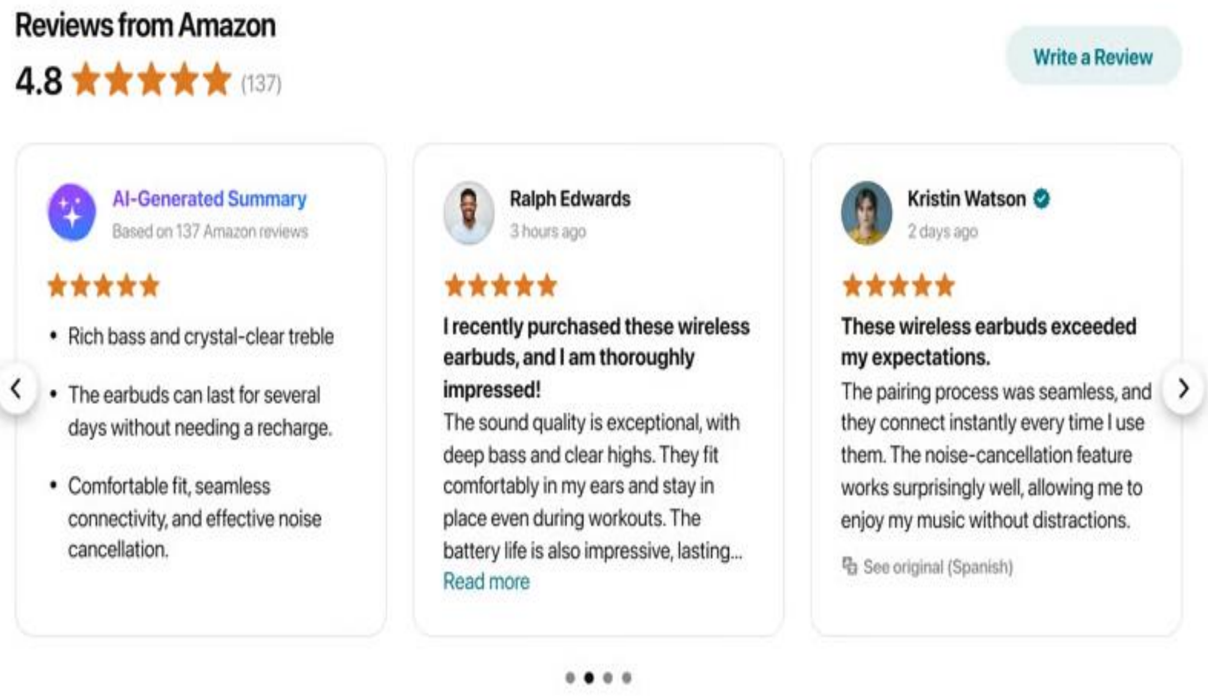
This dataset consists of a few million Amazon customer reviews (input text) and star ratings (output labels)

↓

Sentiment Analysis

↓

How customers react to this product



Dataset

eCommerce behavior data from multi category store dataset

https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store/

Commerce Behavior Dataset offers a broader view of user behavior—including cart removals and session flow—making it ideal for full journey analysis and funnel tracking.

Table data (4 types of events, namely “view”, “cart”, “remove_from_cart” and "purchase" along with product metadata (e.g., category, brand, price).)

Descriptive | **Diagnostic** | **Predictive** | **Prescriptive**

Suggest next products based on customer behavior

Track full user journey and optimize product recommendations

Dataset

User behavior data on Taobao App **dataset**

https://tianchi.aliyun.com/dataset/46

Large-scale user interaction data from the Taobao app, including page views, clicks, cart additions, favorites, and purchases.

Table data (4 types of events: “pv” (view), “cart”, “fav” (favorite), and “buy” (purchase), along with user ID, item ID, category ID, and timestamp).

Descriptive | **Diagnostic** | **Predictive** | **Prescriptive**

Predict likely purchases and personalize product suggestions based on user behavior patterns.

Dataset

MovieLens dataset **dataset**

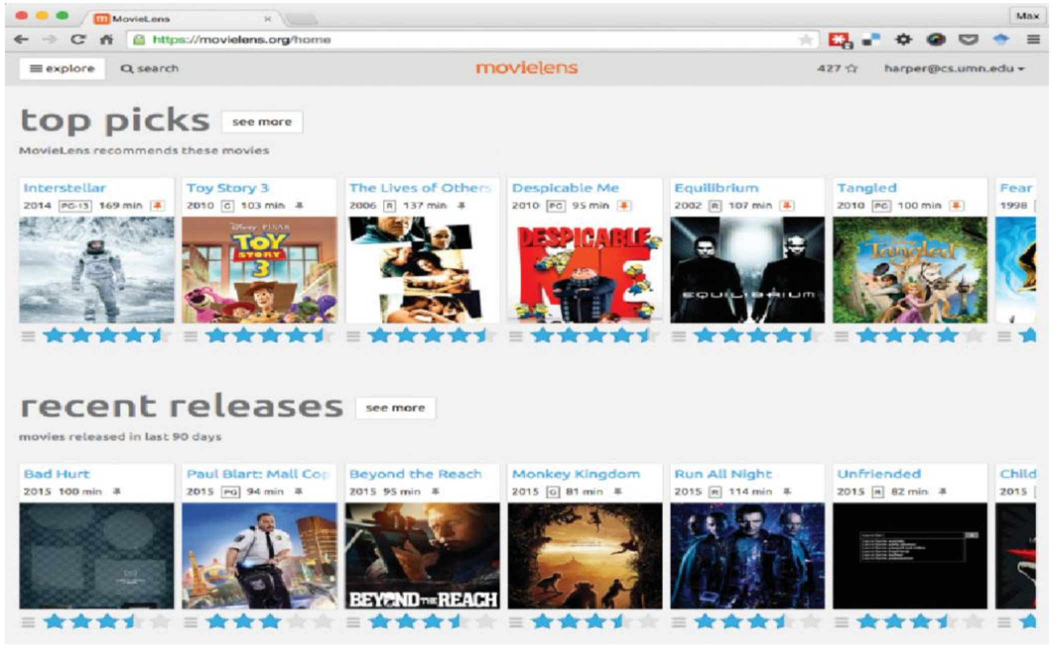
https://www.kaggle.com/datasets/grouplens/movielens-20m-dataset/data/

A benchmark dataset containing 20 million movie ratings from users, widely used for developing and evaluating recommendation systems.

Table data with user ratings (1–5 stars), including movie metadata (title, genre) andoptional tags.

Descriptive **Diagnostic** **Predictive** **Prescriptive**

Build personalized movie recommenders using collaborative filtering or matrix factorization techniques.



Project Guideline

Business Understanding

1 Define Problem and Pain Points

- Defines a Real Business Problem
- Demonstrates User Empathy

2 Define Solutions and Methods

- Presents a Clear, Practical Solution
- Justifies the Approach

3 Define Success Criteria

- Defines Clear Success Metrics
- Aligns with Stakeholder Impact

Data Understanding

1 Describe data chosen

- Describes Data Thoroughly
- Justifies Data Relevance

2 Explore the data chosen

- Applies Diverse Data Exploration Techniques
- Demonstrates Data Understanding

3 Identify key findings and patterns

- Discovers Meaningful Insights
- Links Insights to Business Goals

4 Demo codes for Data Exploration

- Delivers Clean and Reusable Code
- Ensures Reproducibility and Clarity

Data Preparation

1 Describe and explain

Executes and Justifies Data Cleaning

Transforms Data for Business & Modeling Use

2 Demo codes for Data Preparation

Implements Clean, Reusable Code

Ensures Robustness and Reproducibility

Modelling

1 Describe and explain

- Justifies Model Choice Clearly
- Understand pain points & Identify goals

2 Analyze pros and cons

- Evaluates Key Trade-offs
- Balances Business & Technical Needs

3 Describe and explain

- Explains Full Modeling Workflow
- Demonstrates Mastery of Model Lifecycle

4 Demo codes

- Delivers Modular & Reproducible Code
- Ensures Readability & Robustness

Evaluation

1 Evaluate results

- Evaluates Model Performance Thoroughly
- Links Results to Business Goals

2 Review process

- Reviews Workflow Transparently
- Identifies Areas for Improvement

3 Determine next steps

- Outlines Practical Next Steps
- Shows Strategic Foresight

4 Demo codes

- Produces Clear, Modular Code with Evaluation
- Applies Robust Validation Techniques

Deployment

1 Plan deployment

- Outlines Complete Deployment Strategy
- Ensures Scalability and Business Alignment

2 Plan monitoring and maintenance

- Establishes Robust Monitoring Plan
- Prepares for Maintenance and Recovery

3 Review project

- Conducts Thoughtful Project Reflection
- Identifies Lessons Learned for Future Use

Other

1 Reference and Citation

- Uses Consistent Citation Style
- Includes Full References

2 Report with clear format

- Presents Information in a Clear, Structured Format
- Enhances Readability with Visuals and Styling

Conclusion

- **Critical Role of AI**: AI is transforming retail and eCommerce by enhancing personalization, optimizing inventory management, and addressing challenges such as customer engagement and conversion rates, supporting sustainable business practices.
- **Immense Potential**: AI-driven recommendation systems, such as product suggestions, personalized offers, and dynamic pricing, can revolutionize shopping experiences and reduce environmental impact, especially in the growing eCommerce sector in Vietnam.
- Understand the **CRISP-DM (Cross-Industry Standard Process for Data Mining)**
- Skill Development Needs: Expanding expertise in AI models (e.g., CNN, RNN) and data analysis is essential for retailers to effectively implement AI-based solutions like recommendation engines:
    - Knowing the full process to develop solutions like recommendation engines.
    - Gain proficiency in AI techniques such as Machine Learning, Deep Learning, NLP, and Generative AI, including skills like customer segmentation, recommendation algorithms, and fine-tuning models for specific retail tasks.
    - Understand the business dynamics of retail and how data can be leveraged to create personalized shopping experiences that drive customer loyalty and sales growth.

Thank you